PHYC10009 Physics Foundations

Ionising Radiation
Week 11 Lecture 1

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School of Physics
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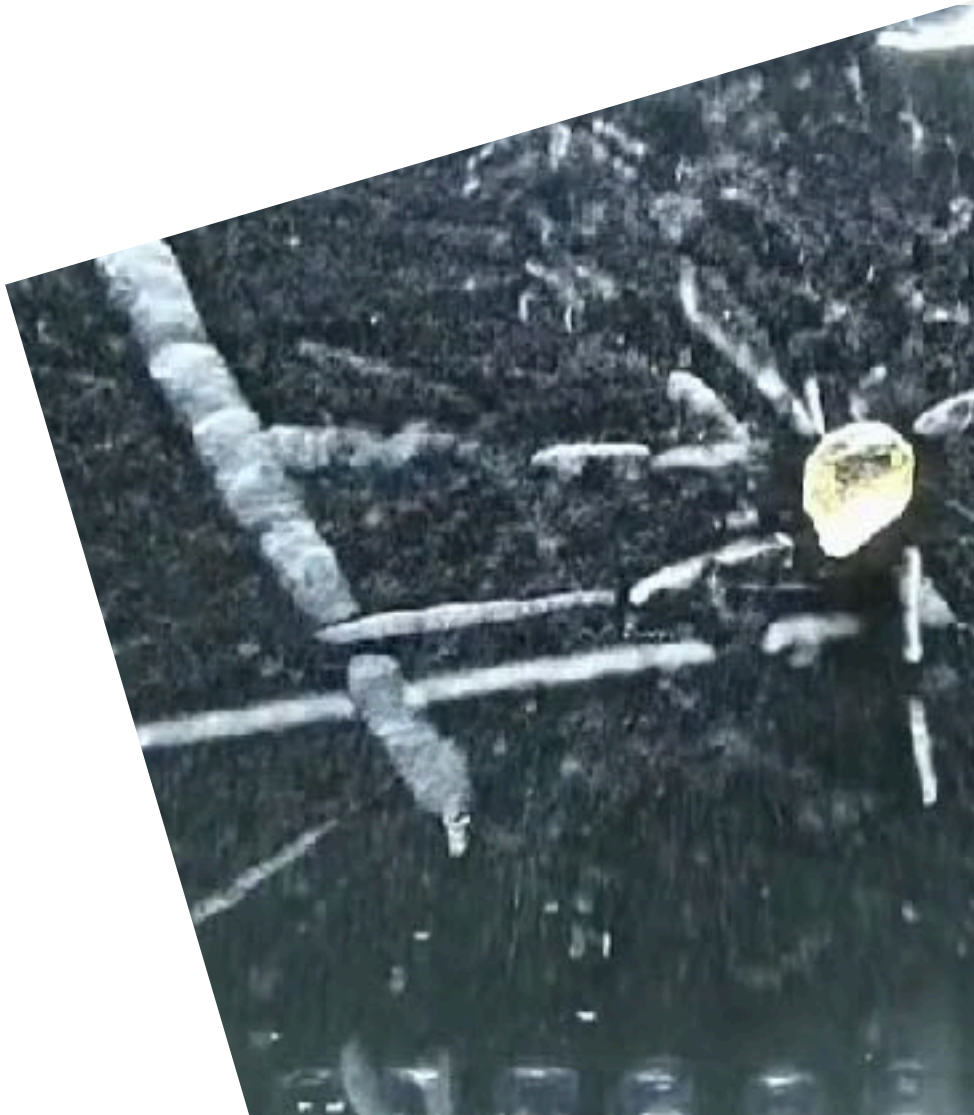
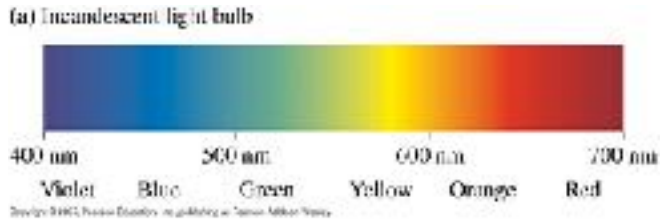


Recap

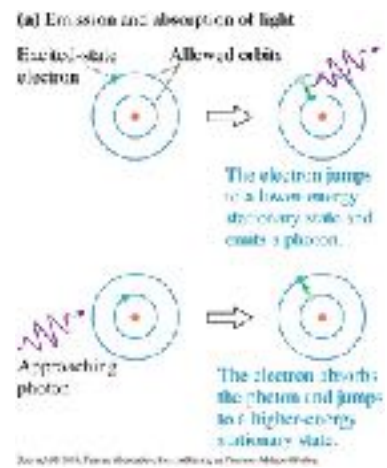


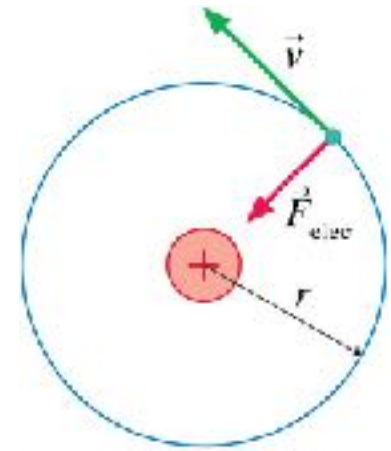
Luminous bodies emit radiation either as continuous spectra or at Discrete wavelengths (frequencies)

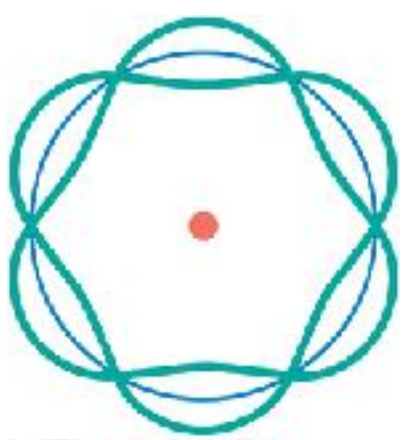
Hydrogen Atom

Both the emission and absorption spectra arise because of the confinement of electrons in atoms.

This gives rise to the quantisation of the energy structure of atoms







Integral multiple of wavelengths requires

$$n\lambda = 2\pi r \qquad r_n = n^2 \frac{h^2 \epsilon_0}{m\pi e^2} \qquad \lambda^2 = \frac{h^2 4\pi r \epsilon_0}{m e^2}$$

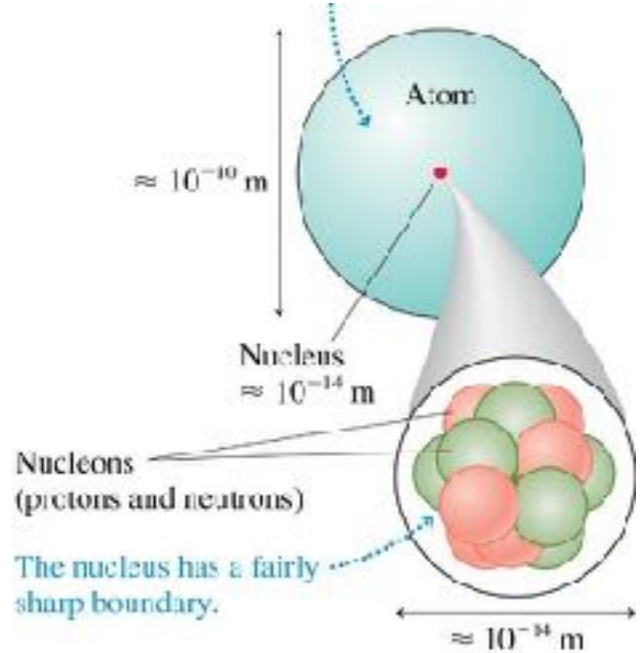
$$E_n = -\frac{13.60}{n^2}\ \text{eV}$$

Recap

Atomic Nucleus

TABLE 30.1 Protons and neutrons

	Proton	Neutron
Number	Z	N
Charge q	$+e$	0
Spin	$\frac{1}{2}$	$\frac{1}{2}$
Mass, in u	1.00728	1.00866



$E = mc^2$

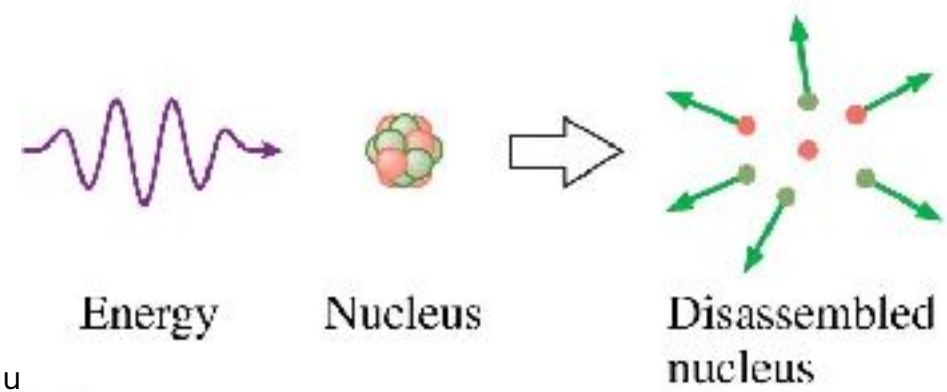
1 u = 1/12th the mass of ^{12}C

$1u = 931.49\ MeV/c^2$

The mass Number A = Z + N is the Total number of Nucleons $^{A}_{Z}X$

Binding Energy

$$B = (Zm_H + Nm_n - m_{atom}) \times (931.49\ MeV/u)$$



Todays lecture

- Nuclear decay
- Alpha, beta and gamma decay
- Radioactivity
- Half-life
- Activity

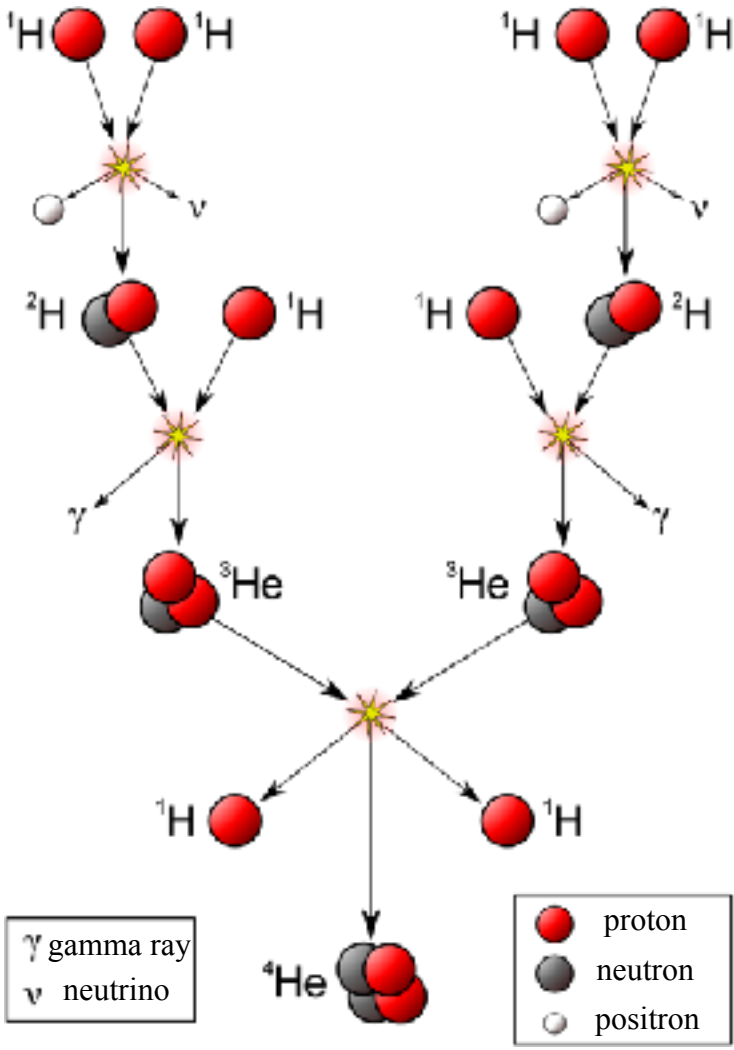
Example of Atomic Mass + Binding Energy

The sun releases energy by converting 4 protons into a Helium Nucleus

It does this via a combination of strong and weak nuclear reactions

The effect is to convert 4 free protons into 2 protons and 2 neutrons bound in a Helium nucleus (an alpha particle).

$$4p \rightarrow {}^{4}He(2p, 2n)$$



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$$4p \rightarrow {}^{4}He(2p, 2n)$$

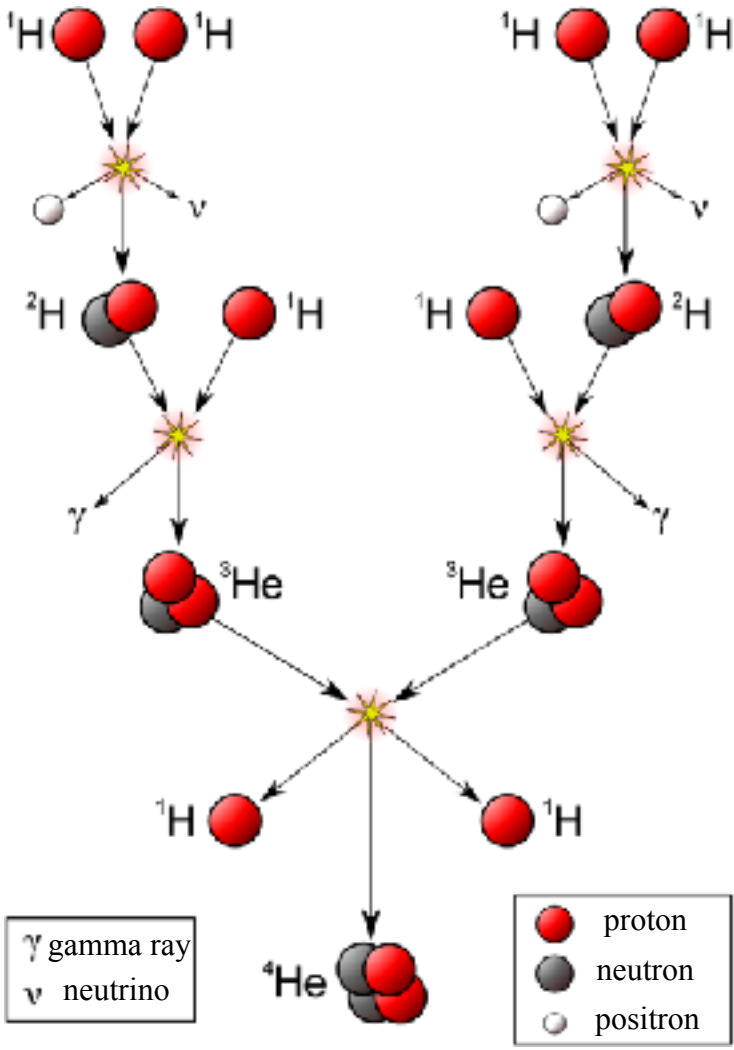
Mass proton = 1.007276u, Mass ^{4}He = 4.001506u
Energy difference =

$$\left(4 \times 1.007276 - 4.001506\right)u = 0.027598\ u$$

Energy difference =

$$0.027598\ u \times 931.49 \frac{MeV}{u} = 25.71\ MeV$$

(1 million times more energy than chemical reaction: $2H + O \rightarrow H_2O$)



Nuclear decay G30-3

Nuclear "decay" is a transition from a high Energy state to lower state

Three types of Radioactive Decay

$\alpha\ decay$: $\quad {}^{A}_{Z}N \rightarrow {}^{A-4}_{Z-2}N' + {}^{4}_{2}He$

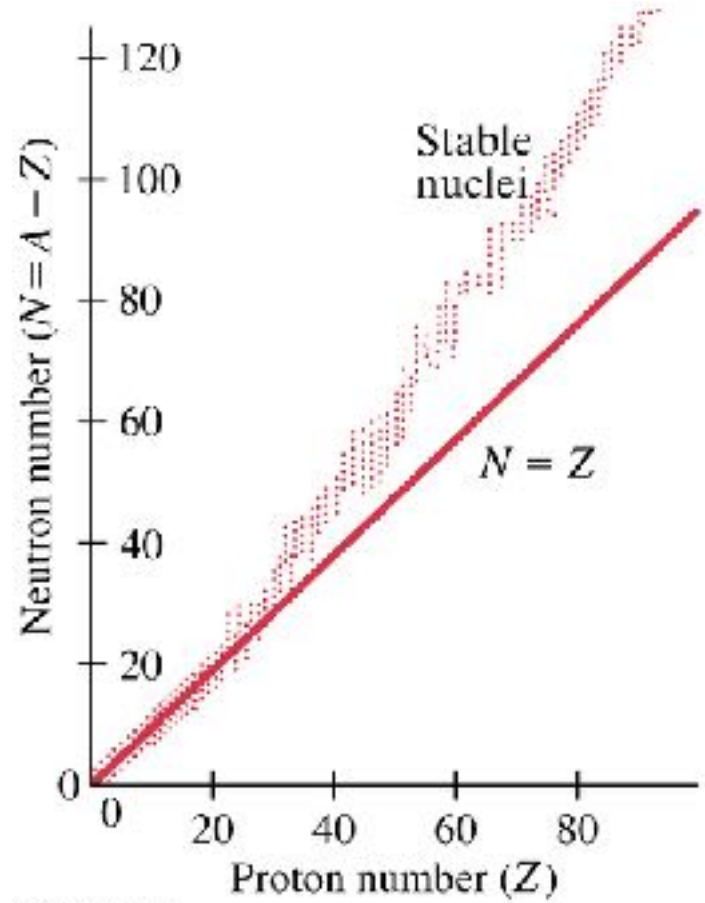
$\beta\ decay$:

$$ {}^{A}_{Z}N \rightarrow {}_{Z+1}^{A}N' + e^{-} + \bar{\nu}_e $$
$$ {}^{A}_{Z}N \rightarrow {}_{Z-1}^{A}N' + e^{+} + \nu_e $$
$$ {}^{A}_{Z}N + e^{-} \rightarrow {}_{Z-1}^{A}N' + \nu_e\ [EC]^{\dagger} $$

$\gamma\ decay$:

$$ {}^{A}_{Z}N^{*} \rightarrow {}^{A}_{Z}N' + \gamma $$

†Electron capture
*Indicates the excited state of a nucleus



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Radioactivity

Radioactive rays were observed to be of three types:

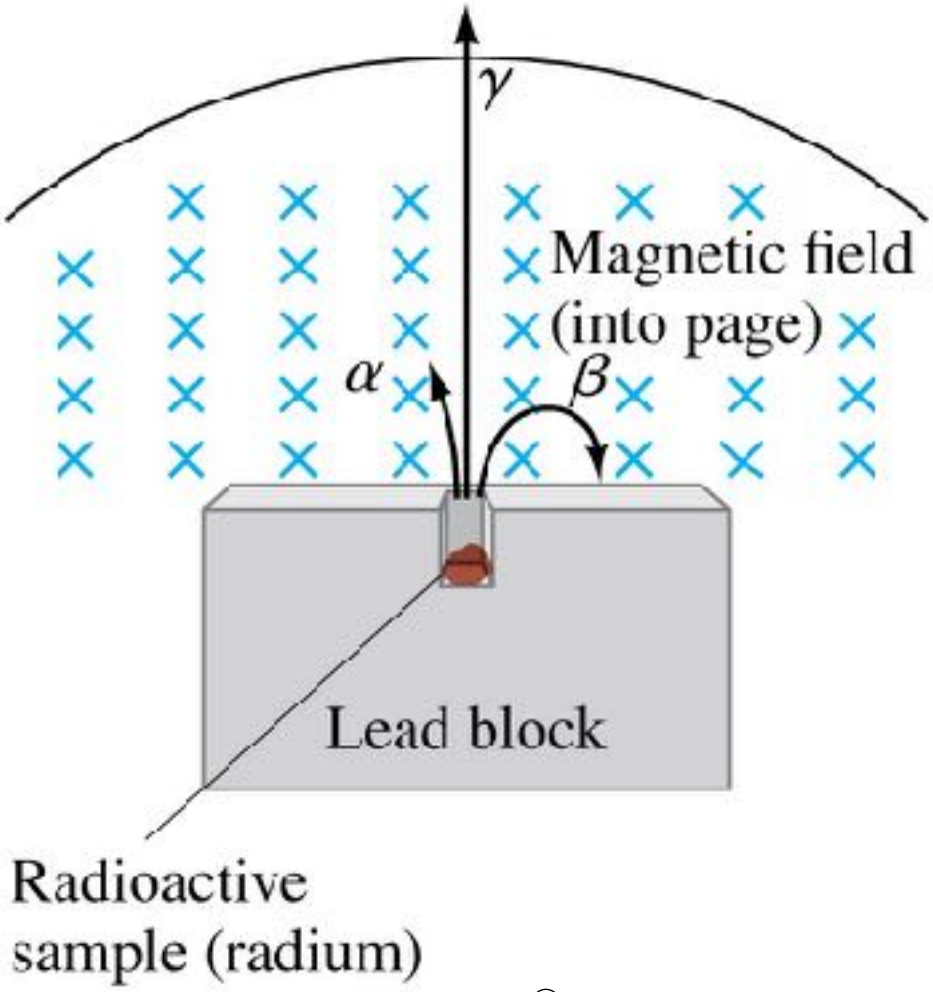
1. Alpha rays, which could barely penetrate a piece of paper
2. Beta rays, which could penetrate 3 mm of aluminum
3. Gamma rays, which could penetrate several centimetres of lead

We now know that alpha rays are helium nuclei, beta rays are electrons, and gamma rays are electromagnetic radiation.

Radioactivity G30.3

Alpha and beta rays are bent in opposite directions in a magnetic field, while gamma rays are not bent at all.

Alpha decay occurs when a very heavy nucleus emits a helium nucleus (2p,2n) to become lighter.



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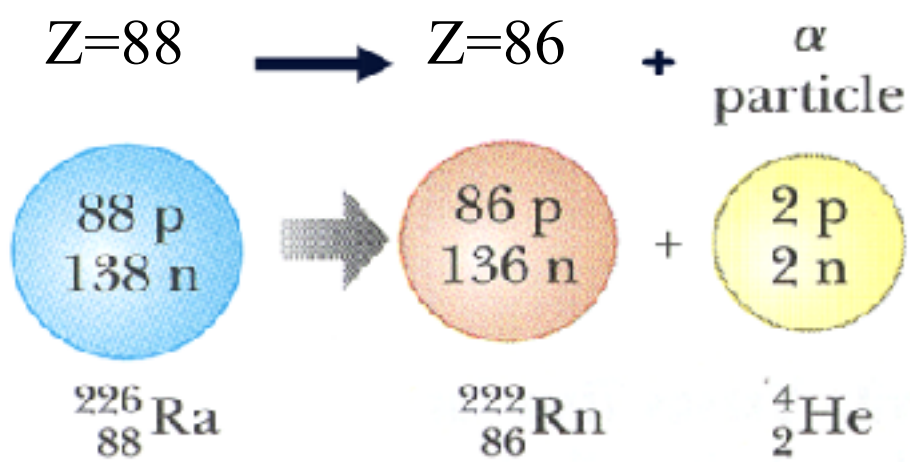
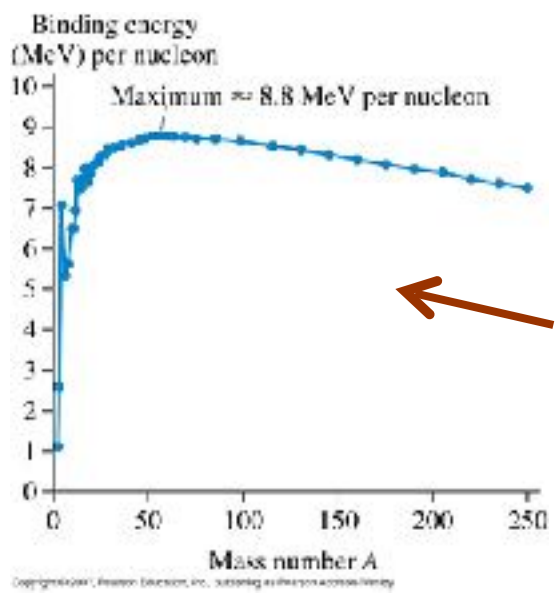
Alpha decay – G30.4

Alpha decay occurs when a heavy nucleus emits a Helium Nucleus to become lighter.



It does this to become more Tightly bound and to move Up the binding Energy/Nucleon curve

$$^{A}_{Z}X \rightarrow ^{A-4}_{Z-2}Y + \alpha$$



Alpha Decay moves 4 units

Beta decay – G30.4

For any given mass number, A, there is one nucleus that is most stable

Beta-decay is the process by which neutrons emit electrons to become protons

Or Where Protons emit positrons to become neutrons.

$$ {}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + e^{-} + \overline{\nu}_{e} $$

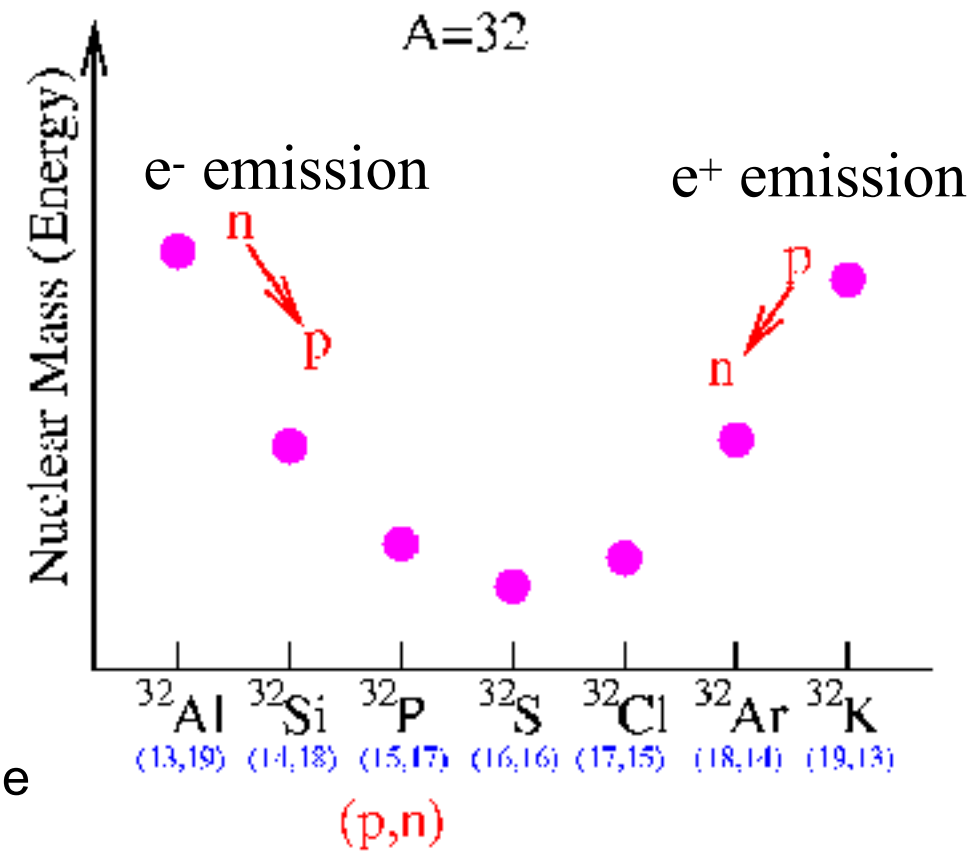
(Electron emission beta-decay)

$$ {}^{A}_{Z}X \rightarrow {}^{A}_{Z-1}Y + e^{+} + \nu_{e} $$

(positron emission beta-decay)

A positron is the anti-particle of an electron. It has the same mass but opposite charge

Positron emission tomography is A powerful diagnostic tool in medicine



Detecting Beta-radiation

Beta-radiation are high energy electrons and positrons (anti-electrons)

$$ {}^{A}_{Z}X \rightarrow {}^{A}_{Z-1}Y + e^{+} + \nu_e $$

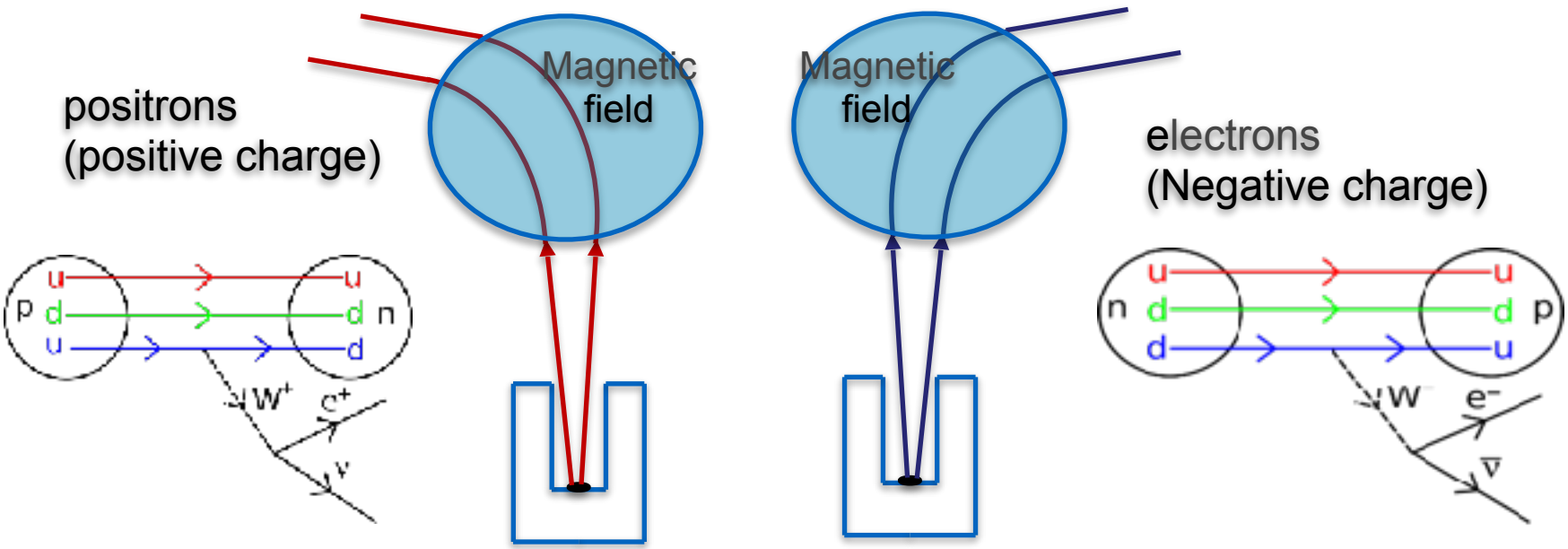
(positron emission beta-decay)

$$ {}^{A}_{Z}X \rightarrow {}^{A}_{Z+1}Y + e^{-} + \overline{\nu}_e $$

(Electron emission beta-decay)

D Deflection of beta-rays

Magnetic Fields deflect moving charged particles
Direction is opposite for sign of electric charge



2011 exam

Question 8 **[2 + 3 + 4 + 3 + 2 + 2 = 16 marks]**

In the days after the Fukushima Nuclear Power accident, an 800 kg cow grazing grass in Fukushima prefecture in Japan ingests radioactive iodine ^{121}I with an activity of 1.00×10^5 Becquerel. The ^{121}I nucleus has 53 protons and 68 neutrons.

(a) ^{121}I decays via beta emission as shown in the decay equation shown.

$$^{121}_{53}I \rightarrow ^{A}_{Z}X + e \quad + \overline{\nu}_e$$

What is the total number of nucleons "A" and protons "Z" of the final state nucleus "X"? (You do not need to determine the chemical element X)

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(a) A=121, Z = 54. (we can work it out from the beta- decay.)

2012 exam

Question 8 **[4 + 4 + 6 + 3 + 3 = 20 marks]**

The average Human Male body has a mass of 70 kg and contains 140 grams of potassium.

(a) Given the isotopic content of ^{40}K is 0.012%, calculate the number of ^{40}K nuclei in the average male body.

$$N_A = 6.02 \times 10^{23} \quad \text{mol}^{-1}$$

(b) ^{40}K undergoes radioactive decay via beta decay and positron emission. In the following show the number of protons and nucleons for the final state nuclei.

(i) $^{40}_{19}K \rightarrow {}^{A}_{Z}Ar + e^{-} + \nu_e$

(ii) $^{40}_{19}K \rightarrow {}^{A}_{Z}Ca + e^{+} + \bar{\nu}_e$

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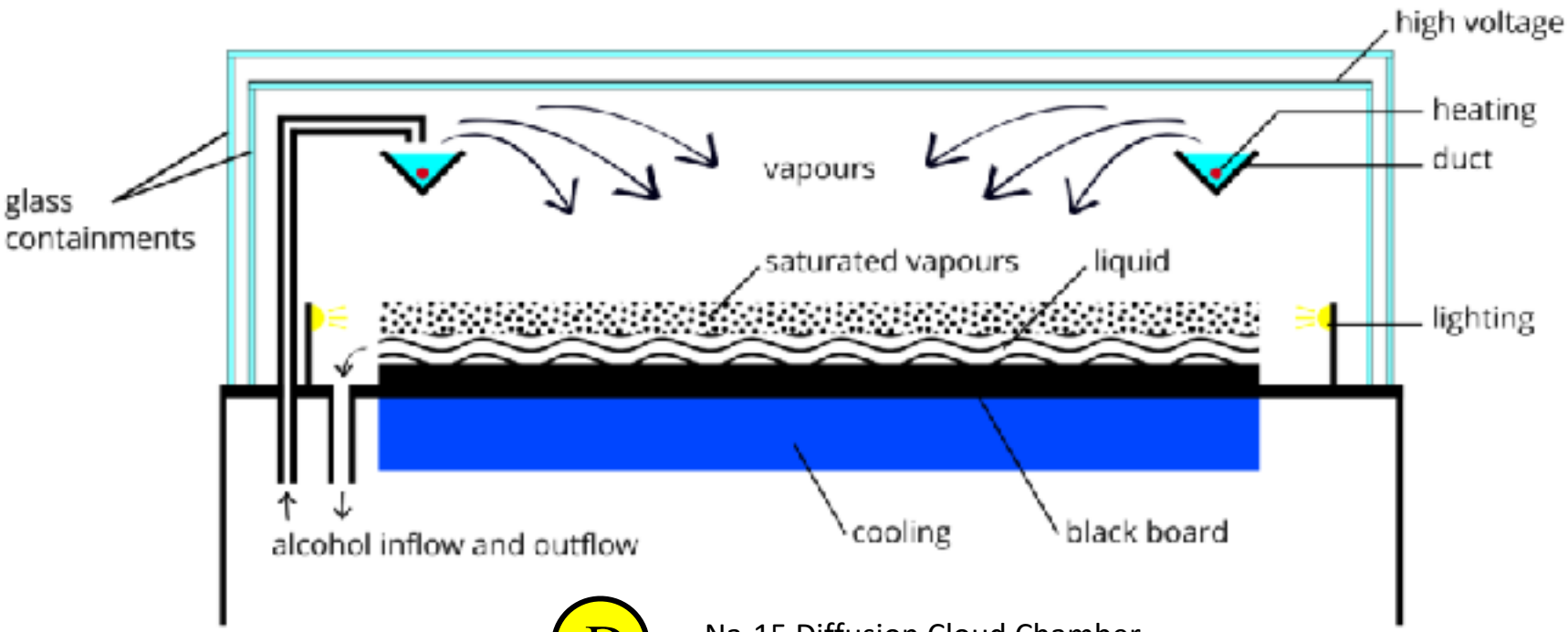
(a) (a) $\frac{140}{39} \times 6 \times 10^{23} \times \left(0.12 \times 10^{-2}\right) = 2.6 \times 10^{20}$ ^{40}K atoms

(b)(i) A= 40, Z=18

(ii) A=40, Z=20

Visualizing radioactivity – Cloud Chamber

A cloud chamber is a device that shows the tracks of alpha, beta (gamma) rays



D Na-15 Diffusion Cloud Chamber

https://www.youtube.com/watch?v=XGNvAEtYZkw

Build your own Cloud Chamber (CERN)

Particles, coming through!
When a particle zips through the cloud chamber, it bumps into atmospheric molecules and knocks off some of their electrons, turning the molecules into charged ions. The atmospheric alcohol is attracted to these ions and clings to them, forming tiny droplets.

The resulting tracks left behind look like the contrails of airplane —long spindly lines marking the particle's path through your cloud chamber.

What you can tell from the tracks?
Many different types of particles might pass through the cloud chamber. You can differentiate between the types of particles based on the tracks they leave behind.

Short, fat tracks
α particle - can be from atmospheric radon.
Bulky, low energy.

Long, straight tracks
Muons from cosmic rays interacting with the upper atmospheric.

Zig zags and curlies
Electrons or positrons.
Light, and bounce around when they hit air molecules.

Forked tracks
Particle decay!

Gamma decay – G30.4

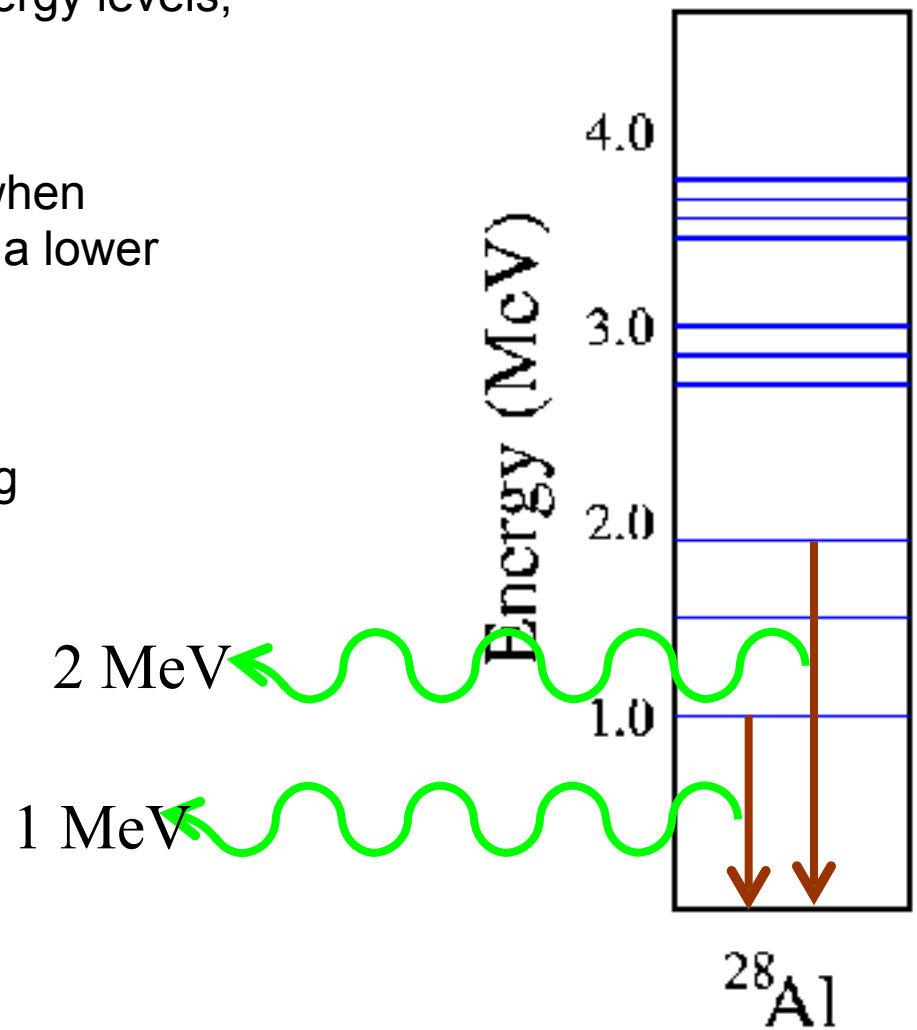
Like atoms, nuclei have well defined energy levels, although their scale is much higher

Gamma-rays are the photons emitted when an excited state of a nucleus decays to a lower Energy level.

Gamma-rays are often emitted following Alpha or beta decays which leave the The nucleus is an excited state.

The energy of the emitted Gamma is energy difference Between the nuclear energy levels

$$ {}^{A}_{Z}X \rightarrow {}^{A}_{Z}X + \gamma $$



Nuclear decay – G30.4

Because the process of emitting radiation results in a transition from a higher energy state to lower energy state, the transition is called *Nuclear Decay*

The speed with which a nucleus decays is called it's *decay rate,* λ

Probability a Nucleus will decay in a small interval of time, Δt, is:

Prob of decay in Δt = λΔt

Then if there are N independent Nuclei the number of decays in Δt is

Number of decays = N x Probability of decay = NλΔt

The rate of change in the number of Nuclei, ΔN per unit time is

$$\frac{\Delta N}{\Delta t} = -\lambda N \quad \text{so} \quad \frac{dN}{dt} = -\lambda N$$

Integrating this gives: $N(t) = N_0 e^{-\lambda t}$ N_0 = Initial number of Nuclei present

Nuclear decay – Half-life

One can define a time constant, the lifetime, τ that is the inverse of the decay rate: $\tau = \frac{1}{\lambda}$, where τ is called the "*lifetime*" then:

$$N(t) = N_0\, e^{\frac{-t}{\tau}}$$

So the number of nuclei present decays with time :

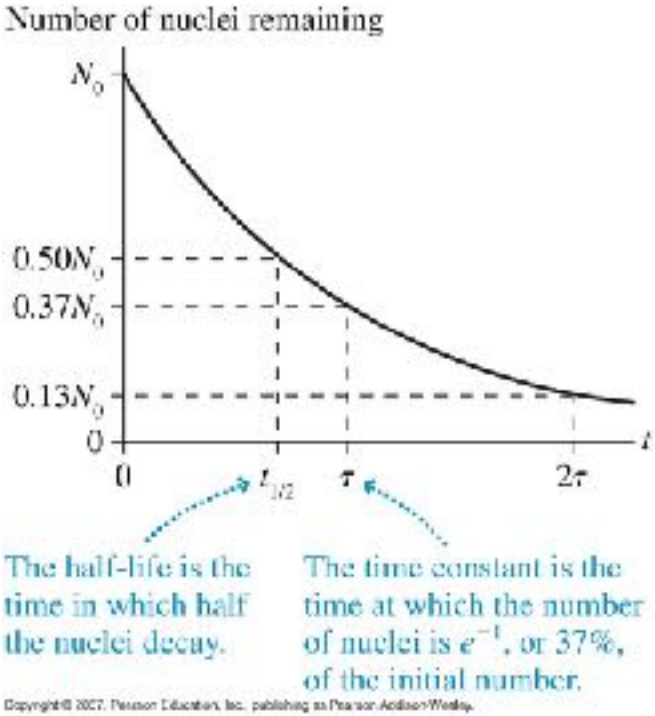
Another common parameter used to quantify nuclear decay is the Half-life $t_{½}$

$t_{1/2}$ is the time required for ½ of nuclei currently present to decay.

$$\frac{N_0}{2} = N_0 e^{-\frac{t_{1/2}}{\tau}} \Rightarrow log_e\left(\frac{1}{2}\right) = -log_e(2) = -\frac{t_{1/2}}{\tau}$$

$$\Rightarrow t_{1/2} = log_e(2)\tau$$

And then: $N(t) = N_0 2^{\frac{-t}{t_{1/2}}}$



Nuclear decay – Half-life

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$$\Rightarrow t_{1/2} = log_e(2)\tau$$

$$=> t_{1/2}=0.693\tau$$

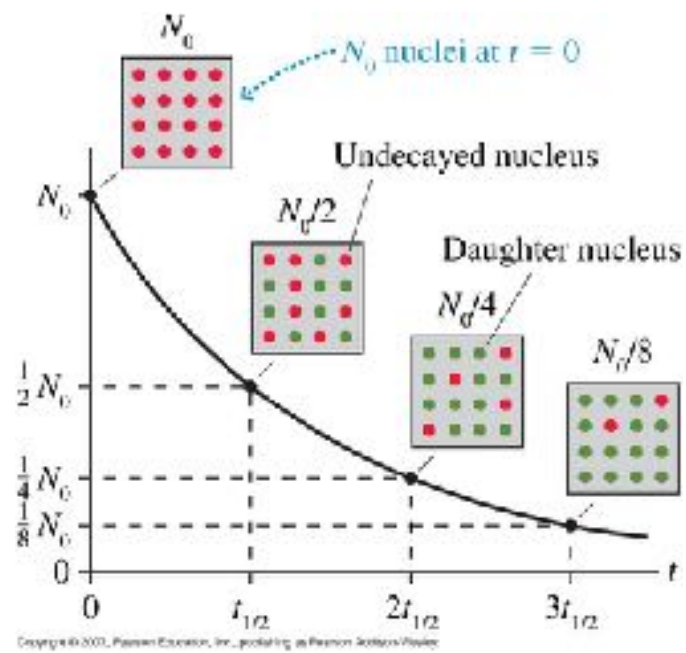
So $t_{1/2}$ is related to the lifetime τ and decay rate λ by

$$\lambda = \frac{1}{\tau} = \frac{0.693}{t_{1/2}}$$

A shorter half-life implies a faster decay rate and vice-versa

Using the half-life gives an alternative expression for the decay of Nuclei

So $N(t=0)=N_0$, $N(t=t_{1/2})=N_0/2$, $N(t=2t_{1/2})=N_0/4$
$N(t=3t_{1/2}) = N_0/8$ etc.



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A sample's half-life is 1 day. What fraction of the original sample will have decayed after 3 days?

$\frac{1}{8}$

$\frac{1}{4}$

$\frac{1}{2}$

$\frac{3}{4}$

$\frac{7}{8}$

All of it

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Activity

The Activity, R, of a radioactive sample is the number of decays per Second of the sample.

From the previous calculations and under the assumption that N does not vary from its initial value N_0 by much :

$$R = N\lambda = = \frac{0.693N}{t_{1/2}}$$

The unit of activity is the becquerel.

1 becquerel = 1 Bq = 1 decay/sec

An older unit, still is widespread use is the curie

1 curie = 1 Ci = 3.7×10^{10} Bq

Activity

Question: The average 70 kg male body contains 140 grams of Potassium. Of this 0.012% is ^{40}K. The half-life of ^{40}K is 1.2 Billion years. How many ^{40}K decay each second in the body? ie What is the Activity of ^{40}K in the body?

First calculate the total number of Potassium nuclei in 140 grams: Use Chemistry!

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Atoms

Number $^{40}K = 2.1\times10^{24}\times1.2\times10^{-4} = 2.6\times10^{20}$ ^{40}K atoms

Half-life in seconds $= 1.2\times10^{9}\times365.25\times24\times3600 = 3.8\times10^{16}$

$$\text{Decay Rate } \lambda = \frac{0.693}{t1/2} = \frac{0.693}{3.8\times10^{16}} = 1.83\times10^{-17}\ s^{-1}$$

Total Decays/s = (Number ^{40}K) x $\lambda = 2.6\times10^{20}\times1.83\times10^{-17}$

= 4700 Bq (decays/s)

So our bodies are (mildly) radioactive

Example: Radio-active dating

It is possible to determine the date of geological and archeological samples By measuring the concentration of naturally occurring radioisotopes.

Organic samples can be dated be measuring the quantity of ^{14}C

^{14}C is produced in the upper-atmosphere via Cosmic rays. It's abundance in the atmosphere is 1.3 parts per trillion. It has a half-life of 5730 years

Living organisms stop absorbing Carbon at death. So the date of death can be determined by the $^{14}C/^{12}C$ ratio in the sample.

Example: A rear molar from a mammoth is dated from the ^{14}C activity. Carbon Is extracted and used to form a a sample of benzene. The ^{14}C activity is 11.5 Counts per minute. A modern benzene sample of the same size has 54.9 Counts per minute. How old is the mammoth?

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$$R(t) = R_0\left(\frac{1}{2}\right)^{\frac{t}{t_{1/2}}} \quad => \quad \frac{R}{R_0} = \left(\frac{1}{2}\right)^{\frac{t}{t_{1/2}}}$$

Solve for t by taking $\log_e$

$$=> \log_e\left(\frac{R}{R_0}\right) = \log_e(0.5)\frac{t}{t_{1/2}} \quad => \quad t = -t_{1/2}\frac{\log_e\left(\frac{R}{R_0}\right)}{\log_e(2)} \quad => \quad t = -\frac{5730}{\log_e(2)}\log\left(\frac{11.5}{54.9}\right) =$$

12900 years

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As a radioactive sample decays:

- The half-life increases
- The half life decreases
- The activity remains the same
- The number of radioactive nuclei increases
- None of the above

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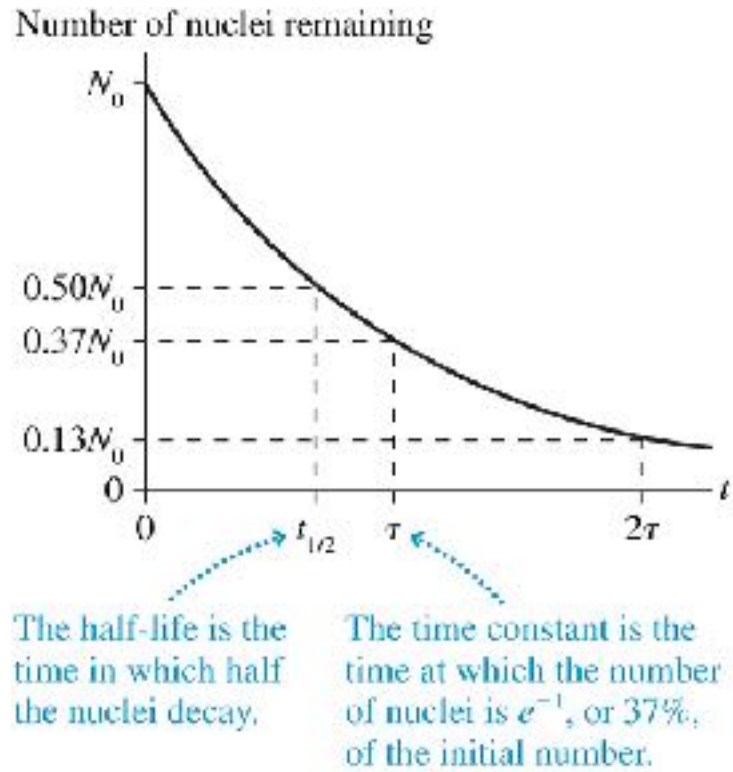
Summary

Nuclear Decay $N(t) = N_0 e^{-\lambda t}$

the lifetime, τ $\tau = \frac{1}{\lambda}$ $N(t) = N_0 e^{\frac{-t}{\tau}}$

Activity, R, $R = \lambda N = \frac{0.693N}{t_{1/2}}$



Alpha decay $^{A}_{Z}X \rightarrow ^{A-4}_{Z-2}Y + \alpha$

Beta Decay

$^{A}_{Z}X \rightarrow ^{A}_{Z+1}Y + e^{-} + \bar{\nu}_e$

$^{A}_{Z}X \rightarrow ^{A}_{Z-1}Y + e^{+} + \nu_e$

Gamma Decay $^{A}_{Z}X \rightarrow ^{A}_{Z}X + \gamma$